TLS_Pie Raspberry Pi Setup Checklist

Use this checklist to set up the Raspberry Pi for the TLS_Pie lidar capture system.

1. Prepare the Pi

Insert the microSD card with Raspberry Pi OS.
Boot the Pi and log in.
Connect the Pi to the network.
Open a terminal.

2. Run the setup script

Run this command:

sudo bash /home/lipi/TLS-Pie/Raspberry\ Pie4/setup_tls_pie_pi.sh

If the project is not yet in /home/lipi/TLS-Pie, copy it there first.

3. Verify the required packages are installed

The script installs:
python3-pip
python3-dev
python3-rpi.gpio
git
tcpdump
openssh-server

4. Verify the project files are present

Make sure these files exist:
/home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPrecord.sh
/home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPbuttons.py
/home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPwaitbutton.py
/home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPstatussignal.py

5. Verify status return wiring

Pi GPIO22 is connected through the level shifter to MicroView D4 (PISTATUS)
Shared ground is connected between Pi and MicroView

6. Test the script manually

Run:

sudo /home/lipi/TLS-Pie/Raspberry\ Pie4/TLS-Pie/VLPrecord.sh eth0

Expected behavior:
the script starts
it waits for the Arduino trigger
it creates a .pcap file in /home/lipi/velodyne
MicroView shows READY before scan
MicroView shows SCANNING during scan
MicroView shows REC when Pi confirms recording

7. If the interface is not eth0

Check the interface name:

ip -br addr

Then run the script with the correct interface name.

8. Enable automatic startup

The setup script creates a startup launcher for the Pi desktop session.

Reboot the Pi:

sudo reboot

9. Confirm the system works

After reboot:
verify the script starts automatically
verify a .pcap file is created
verify the Pi is waiting for the Arduino trigger
verify no PI TIMEOUT message appears unless Pi recording ACK fails

10. Common problems

Wrong network interface name
Missing packages
No common ground between Arduino and Pi
No .pcap file created
Missing GPIO22 -> D4 status return wire

11. Quick recovery commands

If needed, run:

sudo apt update
sudo apt install -y python3-pip python3-dev python3-rpi.gpio git tcpdump openssh-server
sudo pip3 install RPi.GPIO